University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Final Examination, November 2016

Read through the entire examination paper before you start writing so that you can answer the easy questions first. The relative importance of a question is indicated by the marks allocated to it. Please write legibly.

This paper consists of two sections:

Section 1: Advanced Software Design — compulsory
Section 2: Compilers — compulsory

You must answer all questions for both Sections 1 and 2.

Total Marks: 90	Approximate marks per question are shown in brackets.
Time: 3 hours + 10 minutes reading	The use of calculators is permitted.

Please write your answers in two separate books. Use one book for Advanced Software Design, and one for Compilers, and label them as such on the cover page.

Your attention is drawn to the instructions on your answer book on the front and the back. In particular, "**EVERY CANDIDATE MUST enter the book number and the number of each question answered (in the order in which it has been answered)**" on the table provided on the front cover of the first book. Please **do this for the front cover of the different books used for each section.** This is needed so that we can allocate marks and not miss questions. A list of sub-questions is *not* required.

Section 1. Advanced Software Design [45 Marks]

Question 1. **[10]**

a. What problem is being solved by the Repository design pattern and how? [6]

b. The key aim of many patterns is to minimize coupling while maximizing cohesion. Explain what cohesion and coupling are. [4]

Question 2. **[12]**

a. What are the weaknesses and risks of a "code and fix" approach? [2]

b. How do iterative processes accommodate change better than the waterfall approach? [10]

Question 3. **[10]**

Rational Unified Process suggests a "4+1" view of architectures. List and briefly describe all elements of this view.

Question 4. **[13]**

a. Open source projects, just like most large software endeavours, require project management, but due to their nature require some special considerations. Name and explain three issues commonly faced by open source projects. [6]

b. Name and give examples of four types of tools that are commonly needed and used to help manage open source projects. [4]

c. You believe that your source should be freely available for anyone to use regardless of whether they intend to use it for profit or in free software. What license do you use? [3]

Section 2. Compilers [45]

Question 1. **[14]**

a. Indicate two undesirable consequences of disabling optimisations when configuring a compiler, in terms of the resulting executable file. [2]

b. A chip maker has produced a compiler for a particular programming language for their new chip architecture. In time they realise they need to produce a compiler for another programming language to facilitate more development on the new chip architecture. Which phases of the original compiler will have to be rewritten? [4]

c. Given the following grammar, do a leftmost derivation for the string expression "int - int + int", showing every step. [2]

$E \rightarrow T + E$

$E \rightarrow T$

$T \rightarrow int - T$

$T \rightarrow int$

$T \rightarrow (E)$

d. Consider the following Java program containing 3 unique errors identified during compilation. Indicate which line the error is on as indicated on the left and what the error is. For each error also indicate which phase of the compilation process the error is identified - Lexical Analysis, Parsing(Syntactic Analysis) or Semantic Analysis. [6]

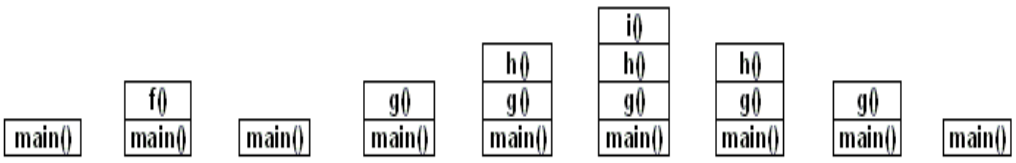
```
1. import java.util.*;
2.
3. class CalculateTotalCost
4. {
5.    public static void main(String[] args)
6.    {
7.       float cost; total;
8.       int items;
9.       char more?;
10.      Scanner kb = new Scanner(System.in);
11.      do
12.      {
13.         System.out.print("Enter item cost:");
14.         cost = kb.nextFloat();
15.         System.out.print("Enter number of items:");
16.         items = kb.nextInt();
17.         total = total + cost * items;
18.         System.out.print("More items(y/n)?:");
19.         more? = kb.next().charAt(0);
20.      }
21.      while(more? == 'y' || more? == 'Y');
22.      System.out.print("Total cost is " + total);
23.   }
24.}
```

Question 2. **[11]**

a. Specify the purpose of an activation record, by also indicating what it contains. [4]

b. Consider the following series of operations on the runtime stack below, indicating the placement of Activation Records for a particular series of subprogram calls. Write skeleton subprogram definitions, with just the subprogram heading and body and specifying which subprograms are called inside of it, in the right order, e.g. [7]

```
sub x() {
    y();
    z();
}
```



Question 3. **[10]**

a. Consider the following IR statements. They define an expression which is assigned to the variable a. Determine the original expression which generated the IR code. [2]

```
MOVE(
  MEM(+(TEMP fp, CONST oa)),
  *(
    CONST 2,
    /(
      CONST 22,
      CONST 7
    )
  )
)
```

b. Explain the IR Code Optimisation techniques constant folding and loop strength reduction. [2]

Consider the following program for Code Block Reorganisation:

```
Start:  Statement0
        Jump B
A:      Statement1
        Jump D
B:      Statement2
C:      Statement3
        Jump A
D:      Statement4
```

c. Separate the code into basic blocks [2]

d. Rearrange the blocks into a trace. [2]

e. Optimise the code by removing redundant jumps. [2]

Question 4. **[10]**

a. At the end of the Instruction Selection process the generated code has "holes". Explain why these holes exist. [1]

b. Use liveness analysis to calculate the live-in and live-out sets for each of the statements in a program below. Show the succ, use, def, in and out sets. Hint: The relevant formulae can be stated as follows: [8]

out[n] = union of in[s] for all successors s of n
in[n] = union of use[n] and (out[n] – def[n])

```
1.  a = b + c;
2.  b = c - d;
3.  d = e / f;
4.  f = a * b;
```

c. During Register Allocation, what causes a "spill"? [1]